

ETV6 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes ETV6 gene fusions from the msk_impact_50k_2026 study, covering 90 fusion events. 64/90 fusions (71.1%) were in-frame. 68/90 fusions (75.6%) retained the PF02198 domain. Domain retention was tested with Fisher's exact test ($p=5.12326e-06$, statistically significant at $\alpha=0.05$). No literature benchmark is configured for ETV6.

1 Results summary

1.1 Fusion partner frequency

Fusion-partner frequency was tabulated across 90 analyzed fusion events, identifying 27 distinct fusion partner genes. The most frequent partner was NTRK3, observed in 55 events.

1.2 Domain retention

PF02198 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 58/64 (90.6%) of in-frame fusions retained the domain, $p=5.12326e-06$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

1.3 Domain disruption

Domain-disruption analysis was skipped: ETV6 has no disruption_required_domains configured; domain-disruption analysis was skipped.

1.4 Cutpoint detection

Cutpoint detection scanned 88 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 155 aa (permutation-corrected $p=0.168317$, not statistically significant at $\alpha=0.05$). This is 32 aa from the nearest configured domain boundary at 123 aa.

1.5 Corroborating confidence statistics

Corroborating confidence statistics were not computed for this run: ETV6 has no group_field configured; confidence-stats analysis was skipped.

1.6 Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

1.7 Exon retention

Exon retention reported: No target exon was supplied in params or GeneConfig.

1.8 Joint partner

Joint partner reported: ETV6 has no gene_pair configured; joint-partner analysis was skipped.

2 Tables

2.1 composite_score composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
NTRK3	55	0.6111111111111112	0.20043213737826426		0.3840220385674932		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.41577810494717626	1
ABCC9	1	0.0111111111111112	0.20043213737826426		1.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.290630525254142	2
ANKS1B	1	0.0111111111111112	0.20043213737826426		1.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.290630525254142	3
ARHGAP26	1	0.0111111111111112	0.20043213737826426		1.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.290630525254142	4
CUX2	1	0.0111111111111112	0.20043213737826426		1.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.290630525254142	5
GALNT18	1	0.0111111111111112	0.20043213737826426		1.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.290630525254142	6
GNB1	1	0.0111111111111112	0.20043213737826426		1.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.290630525254142	7
GRIN2B	1	0.0111111111111112	0.20043213737826426		1.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.290630525254142	8
NOL4	1	0.0111111111111112	0.20043213737826426		1.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.290630525254142	9
PLEKHG7	1	0.0111111111111112	0.20043213737826426		1.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.290630525254142	10
PTPRN2	1	0.0111111111111112	0.20043213737826426		1.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.290630525254142	11
BORCS5	3	0.0333333333333333	0.20043213737826426		0.4747474747474748		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.18760022222383899	12
HCFC1	2	0.0222222222222223	0.20043213737826426		0.38095238095238093		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.16273936879155695	13
CDKN1B	1	0.0111111111111112	0.20043213737826426		0.38095238095238093		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.15797746402965218	14
BCL2L14	3	0.0333333333333333	0.20043213737826426		0.21212121212121215		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.1313231659467827	15
SGO2	2	0.0222222222222223	0.20043213737826426		0.21212121212121215		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.12656126118487793	16
SLIT2	2	0.0222222222222223	0.20043213737826426		0.21212121212121215		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.12656126118487793	17
AEBP2	1	0.0111111111111112	0.20043213737826426		0.21212121212121215		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.12179935642297318	18
APOLD1	1	0.0111111111111112	0.20043213737826426		0.21212121212121215		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.12179935642297318	19
FAM234B	1	0.0111111111111112	0.20043213737826426		0.21212121212121215		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.12179935642297318	20
FGFR1OP2	1	0.0111111111111112	0.20043213737826426		0.21212121212121215		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.12179935642297318	21
IKZF3	1	0.0111111111111112	0.20043213737826426		0.21212121212121215		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.12179935642297318	22
LRP6	1	0.0111111111111112	0.20043213737826426		0.21212121212121215		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.12179935642297318	23
PDE3A	1	0.0111111111111112	0.20043213737826426		0.21212121212121215		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.12179935642297318	24

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty	Components_applicable	Composite_score	Rank
SUFU	1	0.01111111111111112	0.20043213737826426		0.212121212121215		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.12179935642297318	25
CCND2	2	0.02222222222222223	0.20043213737826426				['domain_retention', 'recurrence']	0.10322672911133224	26
PPIL2	2	0.02222222222222223	0.20043213737826426	0.0			['cutpoint_proximity', 'domain_retention', 'recurrence']	0.08110671573033248	27

2.2 cutpoint_detection cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
12	2	66	1	19	0.5757575757575758	0.5433039294306334	0.26495715393209934
155	17	51	9	11	0.4074074074074074	0.10005714167068805	0.9997519077526134
337	67	1	19	1	3.526315789473684	0.40491118077324983	0.3926402309345144
385	68	0	19	1		0.22727272727272727	0.6434526764861874

2.3 domain_retention frame_domain_contingency_table

58	10
6	14

2.4 domain_retention domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained
auto_key_domain	Sterile alpha motif (SAM)/Pointed domain	88	68	0	20	0.0

2.5 frequency Partner_gene_counts

Partner_gene	Event_count
ABCC9	1
AEBP2	1
ANKS1B	1
APOLD1	1
ARHGAP26	1
BCL2L14	3
BORCS5	3

2.6 Per-event results (results.tsv)

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event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_pos	protein_neg	domain_pos	domain_neg	main_retained	main_lost	disrupted_domains	disrupted_domains	annotation	effect_on_fra
EVT-P-0038806-T01-IM6-11	P-0038806-T01-IM6		CUX2-ETV6	CUX2	out-of-frame	three_prime	12008261	4	155	True	lost	0.0	False			Sterile alpha motif (SAM)/Pointed domain		CUX2 (NM_015267) - ETV6 (NM_001987) Rearrangement: c.63+46462:CUX2_c.463+1766:ETV6del	
EVT-P-0007927-T01-IM5-12	P-0007927-T01-IM5		ETV6-ABCC9	ABCC9	out-of-frame	five_prime	12015620	5	155	True	retained	1.0	False			Sterile alpha motif (SAM)/Pointed domain		ETV6 (NM_001987) - ABCC9 (NM_020297) rearrangement: c.464-6738:ETV6_c.3473-739:ABCC9inv	
EVT-P-0060870-T01-IM7-13	P-0060870-T01-IM7		ETV6-AEBP2	AEBP2	out-of-frame	five_prime	12025073	5	337	True	retained	1.0	False			Sterile alpha motif (SAM)/Pointed domain		ETV6 (NM_001987) - AEBP2 (NM_001114176) rearrangement: c.1009+2170:ETV6_c.880-2512:AEBP2del	
EVT-P-0019836-T01-IM6-14	P-0019836-T01-IM6		ETV6-ANKS1B	ANKS1B	in-frame	five_prime	12006878	4	155	True	retained	1.0	False			Sterile alpha motif (SAM)/Pointed domain		ETV6 (NM_001987) - ANKS1B (NM_152788) Rearrangement: c.463+383:ETV6_c.1757-48649:ANKS1Binv	
EVT-P-0027889-T01-IM6-15	P-0027889-T01-IM6		ETV6-APOLD1	APOLD1	out-of-frame	five_prime	12030422	6	337	True	retained	1.0	False			Sterile alpha motif (SAM)/Pointed domain		ETV6:NM_001987) - APOLD1 (NM_001130415) rearrangement: c.1010-6957:ETV6_c.96+30120:APOLD1del	
EVT-P-0025013-T01-IM6-19	P-0025013-T01-IM6		ETV6-BCL2L14	BCL2L14	out-of-frame	three_prime	12025353	5	337	True	lost	0.0	False			Sterile alpha motif (SAM)/Pointed domain		ETV6:APOLD1} BCL2L14 (NM_138722) -ETV6 (NM_001987) Rearrangement: c.945+502:BCL2L14_c.1009+2450:ETV6del	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_pos	domain_pos	main_ref	chr	start	end	disrupted_domains	disrupted_domains	disruption_annotation	fusion_site	effect_on_fusion
EVT-P-0036266-T01-IM6-166	P-0036266-T01-IM6		NOL4-ETV6	NOL4	unknown	three_prime	12015004	5	155	True	lost	0.0	False			Sterile alpha motif (SAM)/Pointed domain		ETV6 (NM_001987) - NOL4 (NM_003787) rearrangement: t(12;18)(p13.2;q12.1)(chr12:g.12015004:15004-15005)x1	NA	
EVT-P-0002978-T03-IM5-167	P-0002978-T03-IM5		NTRK3-ETV6	NTRK3	in-frame	five_prime	12028576	5	337	True	retained	1.0	False			Sterile alpha motif (SAM)/Pointed domain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1 to 5 fused to NTRK3 exons 15 to 20): t(12;15)(p13.2;q25.3)(chr12:g.12028576:1576-1577)x1	NA	
EVT-P-0016507-T02-IM6-168	P-0016507-T02-IM6		NTRK3-ETV6	NTRK3	in-frame	five_prime	12023709	5	337	True	retained	1.0	False			Sterile alpha motif (SAM)/Pointed domain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-14 fused with NTRK3 exons 6-20 including the kinase domain): t(12;15)(p13.2;q25.3)(chr12:g.12023709:1576-1577)x1	NA	
EVT-P-0002978-T06-IM6-169	P-0002978-T06-IM6		NTRK3-ETV6	NTRK3	in-frame	five_prime	12028576	5	337	True	retained	1.0	False			Sterile alpha motif (SAM)/Pointed domain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12028576:1576-1577)x1	NA	
EVT-P-0013196-T03-IM6-170	P-0013196-T03-IM6		NTRK3-ETV6	NTRK3	in-frame	five_prime	12036072	6	337	True	retained	1.0	False			Sterile alpha motif (SAM)/Pointed domain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exon5 fused to NTRK3 exon15): t(12;15)(p13.2;q25.3)(chr12:g.12036072:1576-1577)x1	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	breakpoint_4	breakpoint_5	main_ref	chr1	start	end	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0026390-T01-IM6-185	P-0026390-T01-IM6		NTRK3-ETV6	NTRK3	in-frame	five_prime	12016863	5	155	True	retained	1.0	False	Sterile al-pha motif (SAM)/Pointed domain				ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20): t(12;15)(p13.2;q25.3)(chr12:g.12016863)	NA (NM_001987)	
EVT-P-0026286-T01-IM6-186	P-0026286-T01-IM6		NTRK3-ETV6	NTRK3	in-frame	five_prime	12030395	6	337	True	retained	1.0	False	Sterile al-pha motif (SAM)/Pointed domain				ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused with NTRK3 exons 14-20): t(12;15)(p13.2;q25.3)(chr12:g.12030395)	NA (NM_001987)	
EVT-P-0025987-T01-IM6-187	P-0025987-T01-IM6		NTRK3-ETV6	NTRK3	in-frame	five_prime	12028942	5	337	True	retained	1.0	False	Sterile al-pha motif (SAM)/Pointed domain				ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused with NTRK3 exons 14-20): t(12;15)(p13.2;q25.3)(chr12:g.12028942)	NA (NM_001987)	
EVT-P-0025507-T01-IM6-188	P-0025507-T01-IM6		NTRK3-ETV6	NTRK3	in-frame	five_prime	12024078	5	337	True	retained	1.0	False	Sterile al-pha motif (SAM)/Pointed domain				ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12024078)	NA (NM_001987)	
EVT-P-0023257-T01-IM6-189	P-0023257-T01-IM6		NTRK3-ETV6	NTRK3	in-frame	five_prime	12014998	5	155	True	retained	1.0	False	Sterile al-pha motif (SAM)/Pointed domain				ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1 to 4 fused to NTRK3 exons 14 to 20) : t(12;15)(p13.2;q25.3)(chr12:g.12014998)	NA (NM_001987)	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint	breakpoint	breakpoint	protein	breakpoint	breakpoint	main_ref	chr	start	end	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0010562-T01-IM5-190	P-0010562-T01-IM5		NTRK3-ETV6	NTRK3	in-frame	five_prime	12028903	5	337	True	retained	1.0	False	Sterile al-pha motif (SAM)/Pointed domain				ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 14-20): t(12;15)(p13.2;q25.1)(chr12:g.12028903::chr15:g.885816)	NA	
EVT-P-0021974-T01-IM6-191	P-0021974-T01-IM6		NTRK3-ETV6	NTRK3	in-frame	five_prime	12028417	5	337	True	retained	1.0	False	Sterile al-pha motif (SAM)/Pointed domain				ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12028417::chr15:g.885816)	NA	
EVT-P-0002978-T02-IM5-192	P-0002978-T02-IM5		NTRK3-ETV6	NTRK3	in-frame	five_prime	12028576	5	337	True	retained	1.0	False	Sterile al-pha motif (SAM)/Pointed domain				Fusion: in frame {ETV6:NTRK3} NA NA Protein fusion: in frame (ETV6-NTRK3) ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12028576::chr15:g.885816)	NA	
EVT-P-0006480-T01-IM5-193	P-0006480-T01-IM5		NTRK3-ETV6	NTRK3	in-frame	five_prime	12029548	5	337	True	retained	1.0	False	Sterile al-pha motif (SAM)/Pointed domain				ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12029548::chr15:g.885816)	NA	
EVT-P-0018060-T01-IM6-194	P-0018060-T01-IM6		NTRK3-ETV6	NTRK3	in-frame	five_prime	12037037	6	337	True	retained	1.0	False	Sterile al-pha motif (SAM)/Pointed domain				ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12037037::chr15:g.885816)	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	breakpoint_4	main_ref	chr1	start	end	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0010668-T01-IM5-200	P-0010668-T01-IM5		NTRK3-ETV6	NTRK3	in-frame	five_prime	12029109	5	337	True	retained	1.0	False	Sterile alpha motif (SAM)/Pointed domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12029109: Protein fusion: in frame (ETV6-NTRK3)	NA (NM_001987) - NTRK3 (NM_001012338)	
EVT-P-0014872-T03-IM6-201	P-0014872-T03-IM6		NTRK3-ETV6	NTRK3	in-frame	five_prime	12028946	5	337	True	retained	1.0	False	Sterile alpha motif (SAM)/Pointed domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12028946: Protein fusion: in frame (ETV6-NTRK3)	NA (NM_001987) - NTRK3 (NM_001012338)	
EVT-P-0014872-T02-IM6-202	P-0014872-T02-IM6		NTRK3-ETV6	NTRK3	in-frame	five_prime	12028946	5	337	True	retained	1.0	False	Sterile alpha motif (SAM)/Pointed domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12028946: Protein fusion: in frame (ETV6-NTRK3)	NA (NM_001987) - NTRK3 (NM_001012338)	
EVT-P-0013196-T04-IM6-203	P-0013196-T04-IM6		NTRK3-ETV6	NTRK3	in-frame	five_prime	12036072	6	337	True	retained	1.0	False	Sterile alpha motif (SAM)/Pointed domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exon5 fused to NTRK3 exon15): t(12;15)(p13.2;q25.3)(chr12:g.12036072: Protein fusion: in frame (ETV6-NTRK3)	NA (NM_001987) - NTRK3 (NM_001012338)	
EVT-P-0007772-T01-IM5-204	P-0007772-T01-IM5		NTRK3-ETV6	NTRK3	in-frame	five_prime	12020257	5	155	True	retained	1.0	False	Sterile alpha motif (SAM)/Pointed domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20) : t(12;15)(p13.2;q25.3)(chr12:g.12020257:) Protein fusion: in frame (ETV6-NTRK3)	NA (NM_001987) - NTRK3 (NM_001012338)	

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3 Figures

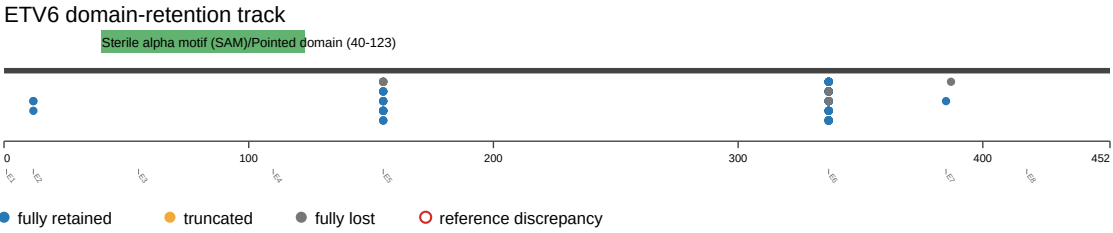


Figure 1: domain retention outliers

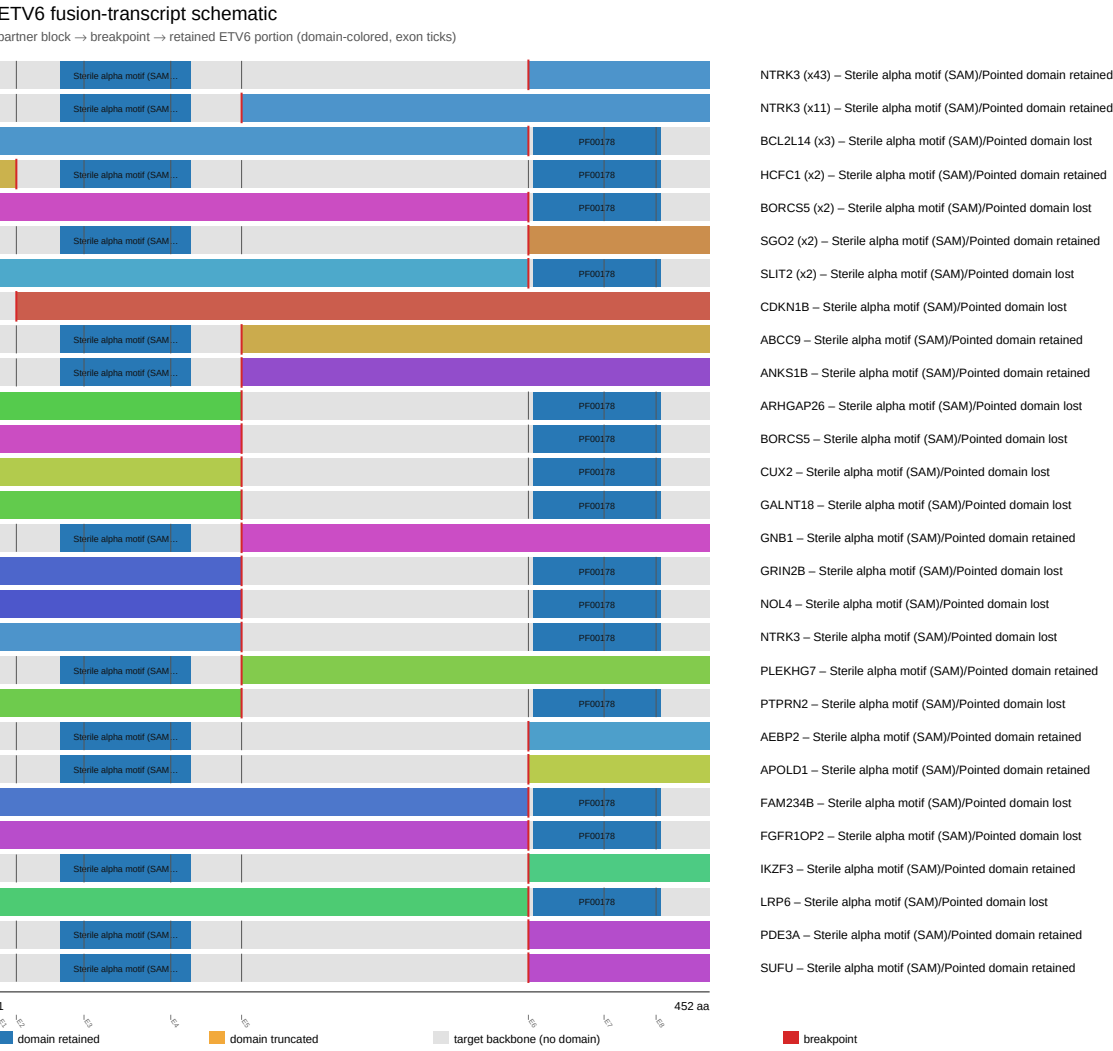


Figure 2: fusion schematic

ETV6 intragenic-deletion schematic

retained N-terminal block → deleted span (plain connector) → resumed C-terminal block

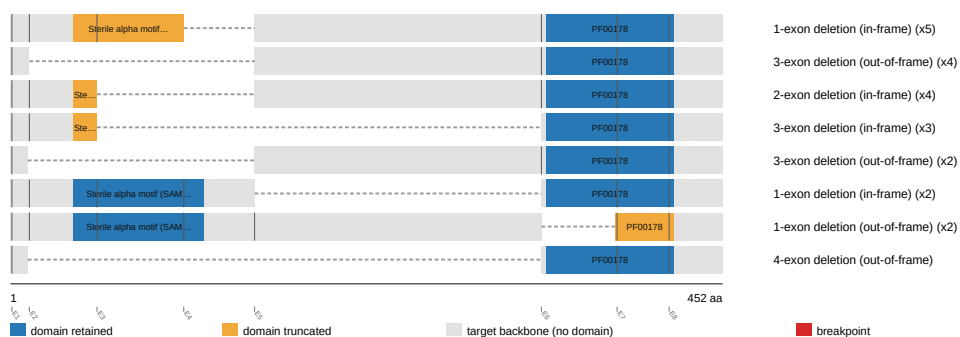


Figure 3: intragenic deletion schematic

Reference vs msk_impact_50k_2026



Figure 4: reference comparison